

1. Administrative & Study Identification

Full Study Title A Study to Evaluate Zilovortamab Vedotin (MK-2140) Combination With Rituximab Plus Cyclophosphamide, Doxorubicin, and Prednisone (R-CHP) Versus Rituximab Plus Cyclophosphamide, Doxorubicin, Vincristine, and Prednisone (R-CHOP) in Participants With Previously Untreated DLBCL (MK-2140-010) **Short Title/Acronym** waveLINE-010 **Protocol Number** MK-2140-010 **NCT Number** NCT06717347 **Sponsor/Company** Merck **Investigational Product** Zilovortamab vedotin (MK-2140) **Phase of Development** PHASE3 **Trial Status** RECRUITING (Currently recruiting participants) **Medical Monitor** James L. Wade III, MD (Principal Investigator) / Gregory Lubiniecki, MD (VP, Oncology Clinical Research)

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate whether zilovortamab vedotin paired with standard treatment (R-CHP) improves progression-free survival (helps participants live longer without the cancer growing or spreading) compared to standard treatment alone (R-CHOP). * **Secondary Objectives:** To evaluate the overall safety, tolerability, complete response at end of treatment (CR at EOT), event-free survival (EFS), and overall survival (OS) of the combination regimen in participants with previously untreated diffuse large B-cell lymphoma (DLBCL).

2.2 Background & Rationale Despite recent advances and the introduction of multi-agent immunochemotherapy (R-CHOP), 5-year survival rates for DLBCL range between 60% and 80%, meaning cure rates remain unsatisfactory. Zilovortamab vedotin is an investigational antibody-drug conjugate (ADC) targeting the receptor tyrosine kinase-like orphan receptor 1 (ROR1), which is overexpressed in multiple hematologic malignancies. Following promising early single-agent and combination activity, this study is being conducted to evaluate if the addition of zilovortamab vedotin to first-line therapy can push the cure rate significantly higher with potentially less toxicity.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (R-CHOP) * **Blinding:** Open-label * **Randomization:** Randomized

3.2 Planned Enrollment & Duration * Target \$\$\$: Approximately 1,046 participants * **Number of Sites:** Global multicenter study * **Estimated Study Start:** 27-Jan-2025 * **Estimated Primary Completion:** 02-Jul-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has histologically confirmed diagnosis of diffuse large B-cell lymphoma (DLBCL), by prior biopsy, based on local testing according to the WHO classification of neoplasms of the hematopoietic and lymphoid tissues. 2. Has positron emission tomography (PET) positive disease at screening, defined as 4 to 5 on the Lugano 5-point scale. 3. Has received no prior treatment for their DLBCL. 4. Has an Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 2 assessed within 7 days before randomization. 5. Has an ejection fraction $\geq 45\%$ as determined by either echocardiogram (ECHO) or multigated acquisition (MUGA). 6. Human immunodeficiency virus (HIV) infected participants must have well controlled HIV on antiretroviral therapy (ART). 7. Participants who are hepatitis B surface antigen (HBsAg) positive are eligible if they have received hepatitis B virus (HBV) antiviral therapy and have undetectable HBV viral load prior to randomization. 8. Participants with history of hepatitis C virus (HCV) infection are eligible if HCV viral load is undetectable at screening.

4.2 Exclusion Criteria 1. Has a history of transformation of indolent disease to DLBCL. 2. Has received a diagnosis of primary mediastinal B-cell lymphoma (PMBCL) or Grey zone lymphoma. 3. Has Ann Arbor Stage I DLBCL. 4. Has clinically significant (i.e., active) cardiovascular disease: cerebral vascular accident/stroke (< 6 months prior to enrollment), myocardial infarction (< 6 months prior to enrollment), unstable angina, congestive heart failure (New York Heart Association Classification Class $\geq$ II), or serious cardiac arrhythmia requiring medication. 5. Has clinically significant pericardial or pleural effusion. 6. Has ongoing Grade > 1 peripheral neuropathy. 7. Has a demyelinating form of Charcot-Marie-Tooth disease. 8. HIV-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 9. Has ongoing corticosteroid therapy. 10. Has received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. Administration of killed vaccines is allowed. 11. Known additional malignancy that is progressing or has required active treatment within the past 2 years. 12. Known active central nervous system (CNS) lymphoma. 13. Has active autoimmune disease that has required systemic treatment in the past 2 years. 14. Has active infection requiring systemic therapy. 15. Has concurrent active HBV (defined as HBsAg positive and detectable HBV DNA) and HCV (defined as anti-HCV antibody positive and detectable HCV ribonucleic acid (RNA)) infection. 16. Has history of allogeneic tissue/solid organ transplant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * Dosage/Frequency: * *Experimental Arm:* Zilovetamab vedotin at 1.75 mg/kg plus cyclophosphamide (trade names: Cytosan, Endoxana) (750 mg/m²), doxorubicin (50 mg/m²), and rituximab (375 mg/m²) via intravenous (IV) infusion on Day 1 of each 21-day cycle. Participants also receive 100 mg prednisone/prednisolone via oral tablet daily on Days 1-5 of each cycle. * *Active Comparator Arm:* R-CHOP (Rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone) administered on Day 1 of each 21-day cycle. * **Route of Administration:** Intravenous (IV) and Oral * **Duration of Treatment:** Up to 6 cycles (21 days per cycle)

5.2 Concomitant & Prohibited Therapy * Permitted: Standard supportive care for chemotherapy-induced side effects (e.g., antiemetics, growth factors). * **Prohibited:** Other investigational agents or concurrent anticancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Biopsy Confirmation, Baseline PET/CT scan
Baseline	Day 0	Randomization, First Dose Administration
Interim	Day 1 of Cycles 1-6	Treatment Infusions, Safety Labs, Efficacy Assessment, AE Monitoring
End of Study	Follow-up	Final Vital Signs, Exit Interview, Disease Progression Monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * Definition: Progression-Free Survival (PFS) – Defined as the time from randomization to the first documented disease progression per Lugano response criteria by blinded independent central review (BICR) or death due to any cause, whichever occurs first. * **Measurement Method:** Radiographic assessment (PET/CT) using the Lugano classification system by BICR.

7.2 Secondary Endpoints * Complete Response at End of Treatment (CR at EOT) * Overall Survival (OS) * Event-free Survival (EFS)

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous monitoring via clinical laboratory tests, physical exams, and patient-reported outcomes at each treatment cycle.

- **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness to the sponsor and applicable regulatory authorities.
 - **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee will review safety and efficacy data at pre-specified intervals.
-

9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) population for all efficacy endpoints; Safety Set for all participants who receive at least one dose of the study treatment.
 - **Sample Size Justification:** $N = 1,046$ participants globally, adequately powered to detect a statistically significant improvement in Progression-Free Survival (PFS) for the experimental arm compared to standard R-CHOP.
 - **Handling of Missing Data:** Standard survival analysis methods (e.g., Kaplan-Meier estimation) with right-censoring for missing data or patients lost to follow-up.
-

10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure data integrity and protocol adherence.
 - **Audit Trail:** Complete electronic Case Report Form (eCRF) tracking with data clarification forms to secure accurate data collection.
-

11. Study Identifiers & Governance

- **Primary ID:** MK-2140-010
 - **Posting ID:** 1131034919335902703
 - **Registry Number:** NCT06717347
 - **Sponsor/Company:** Merck
 - **Study Title:** waveLINE-010
 - **Official Regulatory Title:** A Randomized, Open-Label, Multicenter, Phase 3 Study of Zilovetamab Vedotin (MK-2140) in Combination With R-CHP Versus R-CHOP in Participants With Previously Untreated Diffuse Large B-Cell Lymphoma (DLBCL) (waveLINE-010)
-

12. Clinical Framework (DLBCL Specific)

- **Primary Condition:** Diffuse Large B-Cell Lymphoma (Disease Area: Diffuse Large B-Cell Lymphoma)

- **Preferred UMLS Name:** Diffuse Large B-Cell Lymphoma
 - **Diagnosis Threshold:** Histologically confirmed DLBCL with PET-positive disease (Lugano scale 4-5)
 - **Medical Methodology:** Front-line ADC-based Immunochemotherapy
 - **Optimization Target:** Tumor regression / Complete Response at EOT
-

13. Study Procedures & Methodology

- **Management Pathway:** Standard oncology pathway for first-line DLBCL
 - **Education Protocol (HCP):** Investigator brochure training and protocol site initiation visits
 - **Education Protocol (Patient):** Informed consent process and treatment education materials
 - **Quality Audit Mechanism:** Routine clinical quality assurance audits and source data verification
-

14. Observation & Follow-Up Schedule

- **Total Duration:** Up to Estimated Study Completion Date: March 29, 2032
 - **Onsite Visit Cadence:** Day 1 of every 21-day cycle during treatment, followed by routine surveillance
 - **Remote Monitoring Frequency:** As per standard follow-up protocol intervals
 - **Checklist Review Interval:** Every treatment cycle (Q3W)
-

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	James L. Wade III, MD	_____	[DD- MMM-YYYY]	Clinical Operations Lead	[Name]	_____	[DD-MMM- YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]				